discrete electron charge. Tunnelling cannot raise the electrostatic energy. This is Coulomb blockade. Find the range of plate voltage $V_{AB} = V_A - V_B$ for which single-electron tunnelling is forbidden, that is, the blockade condition.

2. If $V_{AB} = 0.10$ V just permits tunnelling, estimate the capacitance C.

3. For Fig. 1 at fixed voltage V, use the double junction in Fig. 2 to avoid stray capacitance when observing tunnelling. The middle metal layer is the single-electron island. The electrodes are S, D. Its charge is $-ne$, where n is any integer. e is unit charge; MIM capacitances are C_S and C_D.

Show that the charge-dependent (island) electrostatic energy is

$$U_n = \frac{(-ne)^2}{2(C_S + C_D)} .$$

4. In Fig. 3, add a gate to the source (S), drain (D) pair via capacitor C_G. The gate (G) forms the single-electron transistor of Fig. 4. No single electron

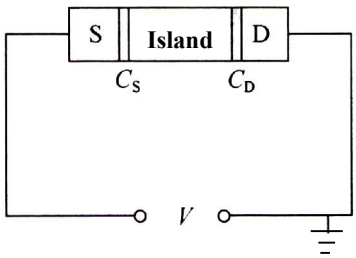


Fig. 3

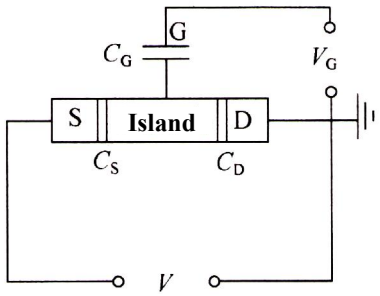


Fig. 4

tunnels between gate and island. At small fixed V, V_G controls n. How does V_G control n? Model V_G as adding the equivalent island charge $q_0 = C_G V_G$. Its energy is about $U_n = (-ne + q_0)^2 / 2C_\Sigma$, with $C_\Sigma = C_S + C_D + C_G$. On the grid (Fig. 5), allow for blockade.

Start with $n = 0$. As $C_G V_G / e$ rises from 0 to 3, draw the island's energy U_n against $C_G V_G$ in bold. Plot U_n vertically; U_n is in

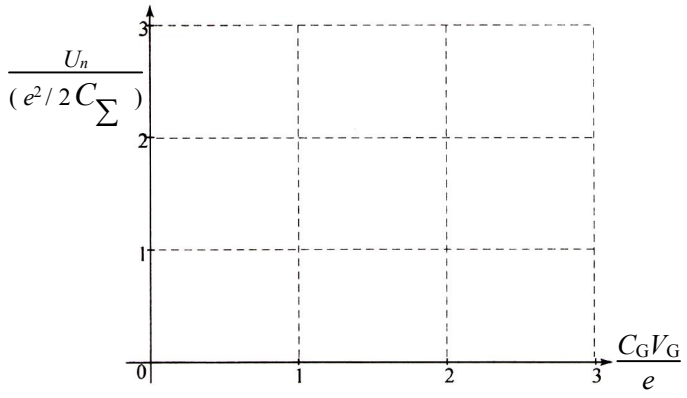


Fig. 5